Aditya **Mohan**

PhD Candidate in Reinforcement Learning at Leibniz University Hannover

a.mohan@ai.uni-hannover.de | amsks.github.io/ | amsks | amsks | amsks96

🤖 *RL researcher with Prof. Dr. Marius Lindauer at Leibniz University Hannover (LUH)*
🤝 *Founding member of autorl.org*
⚡ *Focus on Generalization and State Abstractions in RL, Meta-RL, and AutoRL*

Academic Career

Amazon AWS — *Berlin, Germany*
Research Scientist Intern — *Aug 2025 - Feb 2026*
- **Topic:** Zero-shot Reinforcement Learning for Autoscaling
- **Manager:** Dr. Huibin Shen

Leibniz University Hannover — *Hannover, Germany*
PhD Student — *Since April 2022*
- **Topic:** Generalization in Reinforcement Learning, Meta-Learning and Algorithm Configuration
- **Supervisor:** Prof. Marius Lindauer

Learning and Intelligent Systems Group — *Berlin, Germany*
Research Intern — *Oct 2020 - Dec 2020*
- **Topic:** Plan-Conditioned Policies for Sample-Efficient RL
- **Supervisor:** Ingmar Schubert, Prof. Marc Toussaint

Education

Technical University of Berlin & EURECOM — *Berlin, Germany*
M.Sc. in Autonomous Systems — *2019 - 2021*
- **Thesis:** RL agents that quickly adapt to a partner for Ad-Hoc cooperation in the game of Hanabi (Grade 1.0)
- **Supervisor:** Prof. Klaus Obermayer

Manipal Institute of Technology — *Karnataka, India*
B.Tech. in Electronics and Communication Engineering — *2013 - 2016*
- **Thesis:** Development of Software for Autonomous Driving Support (Grade 1.0)
- **Supervisor:** Dr. S. Bhat M

Publications

Google Scholar **DBLP** **0000-0001-5561-5908**

citations 326 – h-index 8 – i10 index 7

Journal & Conferences

1. Jannis Becktepe, Julian Dierkes, Carolin Benjamins, **Aditya Mohan**, David Salinas, Raghu Rajan, Frank Hutter, Holger Hoos, Marius Lindauer, and Theresa Eimer. *ARLBench: Flexible and Efficient Benchmarking for Hyperparameter Optimization in Reinforcement Learning*. 2026.

2. **Aditya Mohan**, Amy Zhang, and Marius Lindauer. “Structure in Deep Reinforcement Learning: A Survey and Open Problems”. In: *Journal of Artificial Intelligence Research (JAIR)*. **Ranked Q1 According to SCIMAGO**. 2024.

3. Carolin Benjamins, Georgina Cenikj, Ana Nikolij, **Aditya Mohan**, Tome Eftimov, and M. Lindauer. “Instance Selection for Dynamic Algorithm Configuration with Reinforcement Learning: Improving Generalization”. In: *The Genetic and Evolutionary Computation Conference (GECCO)* (2024).

4 Alexander Tornede, Difan Deng, Theresa Eimer, Joseph Giovanelli, **Aditya Mohan**, Tim Ruhkopf, Sarah Segel, Daphne Theodorakopoulos, Tanja Tornede, Henning Wachsmuth, and Marius Lindauer. "AutoML in the Age of Large Language Models: Current Challenges, Future Opportunities and Risks". In: *Transactions on Machine Learning Research (TMLR)* (2024).

5 **Aditya Mohan***, Carolin Benjamins*, Konrad Wienecke, Alexander Dockhorn, and Marius Lindauer. "AutoRL Hyperparameter Landscapes". In: *Proceedings of the Second International Conference on Automated Machine Learning (AutoML Conf)*. 2023.

6 Carolin Benjamins*, Theresa Eimer*, Frederik Schubert, **Aditya Mohan**, Sebastian Döhler, Andre Biedenkapp, Bodo Rosenhahn, Frank Hutter, and Marius Lindauer. "Contextualize Me - The Case for Context in Reinforcement Learning". In: *Transactions on Machine Learning Research (TMLR)* (2023).

7 Mohammed Loni*, **Aditya Mohan***, Mehdi Asadi, and Marius Lindauer. "Learning Activation Functions for Sparse Neural Networks". In: *Proceedings of the Second International Conference on Automated Machine Learning (AutoML Conf)*. 2023.

8 Tim Ruhkopf, **Aditya Mohan**, Difan Deng, Alexander Tornede, Frank Hutter, and Marius Lindauer. "MASIF: Meta-learned Algorithm Selection using Implicit Fidelity Information". In: *Transactions on Machine Learning Research (TMLR)* (2023).

Workshop nad Preprints

1 Jan Malte Töpperwien, **Aditya Mohan**, and Marius Lindauer. "When Are RL Hyperparameters Benign? A Study in Offline Goal-Conditioned RL". In: *arXiv preprint arXiv:2602.05459* (2026).

2 **Aditya Mohan** *, Theresa Eimer *, Carolin Benjamins, Marius Lindauer, and Andre Biedenkapp. "Mighty: A Comprehensive Tool for studying Generalization, Meta-RL and AutoRL". In: *Eighteenth European Workshop on Reinforcement Learning*. 2025.

3 Dennis Jabs *, **Aditya Mohan** *, and Marius Lindauer. "Moments Matter: Stabilizing Policy Optimization using Return Distributions". In: *Multi-disciplinary workshoperence on Reinforcement Learning and Decision Making (RLDM)*. 2025.

4 **Aditya Mohan** and Marius Lindauer. "Towards Enhancing Representations in Reinforcement Learning using Relational Structure". In: *17th European Workshop on Reinforcement Learning (EWRL)*. 2024.

5 Jannis Becktepe, Julian Dierkes, Carolin Benjamins, **Aditya Mohan**, David Salinas, Raghu Rajan, Frank Hutter, Holger Hoos, Marius Lindauer, and Theresa Eimer. "ARLBench: Flexible and Efficient Benchmarking for Hyperparameter Optimization in Reinforcement Learning". In: *17th European Workshop on Reinforcement Learning (EWRL)*. 2024.

6 **Aditya Mohan**, Amy Zhang, and Marius Lindauer. "A Patterns Framework for Incorporating Structure in Deep Reinforcement Learning". In: *16th European Workshop on Reinforcement Learning (EWRL)*. 2023.

7 **Aditya Mohan**, Tim Ruhkopf, and Marius Lindauer. "Towards Meta-learned Algorithm Selection using Implicit Fidelity Information". In: *ICML 2022 Workshop Adaptive Experimental Design and Active Learning in the Real World (ReALML)*. 2022.

Honors & Awards

Dec 2024	**Featured Publication in the Binare Magazine for excellent AI research**, https://www.l3s.de/magazine/	*Hannover, Germany*
Oct 2024	**Best Publication Award – L3S**, For our JAIR paper "Structure in Deep Reinforcement Learning: A Survey and Open Problems"	*Hannover, Germany*

Funding, Fellowships & Scholarships

2025	**DAAD IFI - Internationale Forschungsaufenthalte für Informatikerinnen & Informatiker**, €10,000
2020	**EIT Digital Merit Scholarships**, €4,500
2017	**DAAD – Praktikenplätze für Ausländishce Studierende der Natur- und Ingineurwissenschaften sowie der Land- und Fortwirtschaft (IAESTE)**, €3,200

Organising

AutoML School 2024 — *Hannover, Germany*
Organiser — *Sept 2024*

Keynote by Chelsea Finn: Meta-Learning for Education — *Baltimore, MA, USA*
Moderator — *Jul 2022*

DAC4AutoML Competition at AutoML Conference 2022 — *Baltimore, MA, USA*
Organiser — *Jul 2022*

Public Outreach

Dec 2024	**CAIRNE Rising Researchers Network**, Organizing collaboration between Industry and Academia
Oct 2024	**Winner – Haus der Wissensschaft Science Slam**, Popular Science Communication Format
Sep 2024	**Meet the Scientist**, Interacting with School Students about AI
Sep 2024	**AutoML Conf Non-Traditional Content**, Musical Parody ”On the Dangers of Grid Search”
July 2024	**AI Grid Science Slam**, Popular Science Communication Format, Second Place in Audience Voting
Nov 2023	**Nacht der Wissenschaft**, University Science Night: RL for all ages

Research Visits

May 2026 - Aug 2026	**Prof. Amy Zhang at University of Texas at Austin**, Zero-shot Reinforcement Learning with structured representations	*Texas, Germany*
Feb 2025 - Apr 2025	**Prof. Georg Martius at University of Tübingen**, Representation Learning from action-free video data for Downstream Control and RL	*Tübingen, Germany*
Jul 2023, Jul 2024	**Dr. Tome Eftimov at Jožef Stefan Institute**, Dynamic Algorithm Configuration using RL	*Ljubljana, Slovenia*

Committees

Since 2024	**Member**, Appointment Committee for the W3 Databases Professorship at the Faculty of Electrical Engineering and Computer Science	*Leibniz Universität Hannover*

Reviewing

JAIR (2024), NeurIPS (2023), ICML (2022), AutoML Conf (2024, 2023, 2022), EWRL (2024, 2023), ICLR (2024, 2023, 2022), ICLR Tiny Papers (2024, 2023), RLC (2026), TMLR

Teaching

Oct 2022 - Aug 2025	**Reinforcement Learning (3x)**, Graduate lecture: Creation and grading of exercises & final project. Teaching concepts for virtual, hybrid, and in-person versions of the course.*Teaching evaluation: (1.8)*
Oct 2024 - Feb 2025	**Project: Reinforcement Learning and Robotics**, Graduate level project course: Course development & Co-Lecturer
Apr 2022 - Jul 2022	**Reinforcement Learning Seminar**, Graduate lecture: Creation and grading of exercises & final project. Teaching concepts for virtual, hybrid, and in-person versions of the course

Mentorship

Apr 2025 - Oct 2025	**Johs Falkenberg (Bachelor Thesis)**, Dynamics Generalization in Offline Goal-Conditioned RL
Oct 2024 - Oct 2025	**Jan Malte Töpperwein (ML Project, Master Thesis)**, Hyperparameter Landscapes of Self-supervised Reinforcement Learning – **PHD Student at LUH Hannover**
Since Apr 2024	**Tim Grunwald (ML Project, M.Sc Thesis)**, Prior-fitted Reinforcement Learning for Algorithm Selection
Feb 2024 - Aug 2024	**Dennis Jabs (M.Sc Thesis)**, Improving Policy Optimization Using Return Landscapes – **Accepted at RLDM 2025**
Feb 2024 - Aug 2024	**Dimitrios Timoleon (M.Sc Thesis)**, Enhancing Reinforcement Learning using Transformer-based Self-Predictive Representations
Since Jun 2023	**Wladislaw Petscherski (B.Sc Thesis, ML Project)**, Activation Functions for Transfer-learning in Reinforcement Learning
Jan 2023 - May 2023	**Lingxiao Kong (M.Sc Thesis)**, Impact of Hyperparameters on Sim2Real Transfer in Reinforcement Learning – **PHD Student at Fraunhofer FIT**
Oct 2022 - Mar 2023	**Konrad Wienecke (M.Sc Thesis)**, Dynamic Hyperparameter Landscapes in Reinforcement Learning – **Published at AutoML Conf 2023**